KojoBench2 — Model Evaluation Report

Qwen 2.5 Coder 7B · 75 Tasks · Zero-Shot Code Generation · Kojo (Scala) turtle graphics

1. Overview

Metric	Qwen 2.5 Coder 7B	Claude Sonnet 4.6
Visual Pass Rate (NSS >= 65%)	**8 / 75 (10.7%)**	**39 / 75 (52.0%)**
Average NSS	36.4%	66.6%
Average KCSS (code quality)	77.1%	81.3%
Render Failures (CRASH)	6 / 75	0 / 75
Tasks using // comments	66 / 75	—
hop(x,y) two-arg hallucination	5 tasks → CRASH	0
Missing import (sqrt)	1 task → CRASH	0

NSS = Normalised Shape Similarity (0.7 × chamfer score + 0.3 × edge correlation). KCSS = Kojo Code Style Score (0.4 × structure + 0.3 × idioms + 0.3 × simplicity). CRASH = Scala compilation failure; no NSS computed.

2. Tasks Passed (NSS >= 65%)

ID	Description	NSS	Category
T1	Circle	**100%**	Primitive
T3	Square	**100%**	Primitive
T4	Pentagon, flat bottom	**66%**	Regular polygon
T11	Rectangle (longer sides horizontal)	**97%**	Primitive
T19	Octagon, flat top and bottom	**71%**	Regular polygon
T29	Square with diagonal top-left to bottom-right	**100%**	Primitive+
T54	Six regular hexagons arranged in a ring	**67%**	Structured repeat
T74	Square, side length 200 (variant B)	**80%**	Size-specific

T26 scored exactly 65% and fell just below the pass threshold. All 8 passes are simple single-shape or regular-polygon tasks solvable with a single repeat loop. No composite, no arc-sequencing, no orientation-specific task passed.

3. Failure Taxonomy

Category	Count	Exemplar Tasks	Avg NSS	Root Cause
CRASH — hop(x,y) two-arg	5	T32, T44, T46, T61, T62	—	Kojo hop(n) takes one arg; Qwen invents Python-style hop(x,y)
CRASH — missing import	1	T73	—	sqrt(3) used without scala.math import
Orientation / mirroring	5	T2, T12, T16, T17, T30	5%	Correct shape type; wrong rotation or reflection
Arc / semicircle sequencing	12	T9,T21,T35,T40,T41,T42,T43,T65,T67,T68,T69,T71	22%	right(angle,r) syntax known; post-arc turtle state ignored
Multi-shape composition	14	T6,T7,T22,T25,T31,T45,T47,T56,T57,T60,T63,T64,T70,T75	[illegible]	Turtle not repositioned between shapes; second shape drawn at wrong position
Nesting / concentric	6	T20,T39,T48,T55,T58,T59	51%	Closest to passing; scale or offset slightly wrong
Size-specific (GT mismatch)	3	T34, T53, T72	28%	GT code for these tasks draws a different shape than the query describes
Other composite / polygon	15	T5,T8,T10,T13,T14,T15,T18,T24,T27,T28,T33,T36,T38,T50,T51	34%	Shape partially correct; composition or count wrong

4. Arc / Semicircle vs. Non-Arc — NSS Comparison

Group	Task Count	Avg NSS	Interpretation

Arc / semicircle / circle tasks	28	36.5%	Syntax correctly recalled; sequencing fails identically to non-arc tasks
Non-arc tasks	47	36.4%	Same failure profile — the gap is 0.1 percentage points
All tasks (excluding 6 crashes)	69	39.6%	Crashes excluded from NSS average
All 75 tasks	75	36.4%	Crashes counted as 0% NSS

Arc tasks and non-arc tasks are statistically indistinguishable by NSS (36.5% vs 36.4%). Arcs are not a special failure mode — the underlying cause is turtle state tracking, which breaks for any sequential multi-step drawing regardless of whether arcs are involved.

5. Core Diagnosis

Failure Mode	Detail
Spatial state tracking	Qwen does not propagate turtle position and heading across drawing steps. Each subsequent shape is drawn as if the turtle
Allocentric orientation	Orientation constraints ('pointing down', 'diagonal from top-right') are ignored. The shape type is pattern-matched but not rota
API surface hallucination	hop(x, y) is invented from Python Turtle / Processing conventions. Kojo's hop(n) accepts only a single distance argument. 66
Declarative vs. procedural gap	Arc syntax right(angle, radius) is correctly recalled (declarative memory), but the model cannot simulate what the call does to
KCSS / NSS divergence	Average KCSS 77.1% vs average NSS 36.4% — a 40.7 point gap. The model writes syntactically plausible, structurally valid
Size-specific inconsistency	T72 and T74 share the query 'Draw a square with side length 200' yet score 37% and 80% respectively. This reflects a GT in

6. Full Task Score Table (all 75 tasks)

ID	Description	NSS	Result	Category
T1	Circle	100%	PASS	Primitive
T2	Triangle pointing up	2%	FAIL	Triangle / orientation
T3	Square	100%	PASS	Primitive
T4	Pentagon, flat bottom	66%	PASS	Regular polygon
T5	Triangle with smaller triangles at each corner	25%	FAIL	Composite
T6	Two side-by-side upward triangles sharing a corner	28%	FAIL	Composite / triangle
T7	Square + right-pointing triangle on right side	24%	FAIL	Composite
T8	Four-pointed star (square centre + 4 triangles)	48%	FAIL	Star / composite
T9	S-curve (top curves right, bottom curves left)	29%	FAIL	Arc / curve
T10	Circle inscribed in a square (touches midpoints)	45%	FAIL	Arc + composite
T11	Rectangle (longer sides horizontal)	97%	PASS	Primitive
T12	Right-angle triangle (two sides axis-aligned)	3%	FAIL	Triangle / orientation
T13	Hexagon, flat top and bottom	20%	FAIL	Regular polygon / orientation
T14	Five-pointed star	22%	FAIL	Star
T15	Two rectangles joined at corner forming an L	24%	FAIL	Composite
T16	Square balanced on one corner (diamond)	0%	FAIL	Orientation
T17	Equilateral triangle pointing downward	3%	FAIL	Triangle / orientation
T18	Rightward arrow (rect body + triangle head)	8%	FAIL	Composite / orientation
T19	Octagon, flat top and bottom	71%	PASS	Regular polygon
T20	Square centred inside larger square, sides parallel	39%	FAIL	Composite / nesting
T21	Semicircle, flat side at bottom	27%	FAIL	Arc / semicircle
T22	Wide rect + narrower rect on top forming a T	13%	FAIL	Composite
T23	Staircase of three equal steps, lower-left to upper-right	0%	FAIL	Staircase / composite

T24	Trapezoid (parallel top and bottom, top shorter)	16%	FAIL	Polygon / orientation
T25	Three equal upward triangles side by side	36%	FAIL	Composite / triangle
T26	Three adjacent squares sharing a base in a rectangle	65%	FAIL*	Composite
T27	Square inscribed inside a circle	40%	FAIL	Arc + composite
T28	Square with both diagonals drawn	45%	FAIL	Composite
T29	Square with diagonal top-left to bottom-right	100%	PASS	Primitive+
T30	Square with diagonal top-right to bottom-left	15%	FAIL	Orientation
T31	Two squares connected at exactly one corner	0%	FAIL	Composite
T32	Two circles stacked + vertical diameter of upper	CRASH	CRASH	Arc / hop(x,y)
T33	Two overlapping squares	33%	FAIL	Composite
T34	Square, side length 80	18%	FAIL	Size-specific
T35	Semicircle on top of equilateral triangle (shared base)	26%	FAIL	Arc + composite
T36	Two circles touching at one point, larger on left	39%	FAIL	Arc / multi-circle
T37	Rectangle with rounded corners	3%	FAIL	Arc / rounded rect
T38	Three circles of different radii sharing one point	48%	FAIL	Arc / multi-circle
T39	Three concentric circles	57%	FAIL	Arc / concentric
T40	Semicircle, radius 60	0%	FAIL	Arc / semicircle
T41	Two semicircles and one triangle	30%	FAIL	Arc + composite
T42	Triangle with two semicircles at base	26%	FAIL	Arc + composite
T43	Square with four semicircles at each corner	42%	FAIL	Arc + composite
T44	Three overlapping semicircles	CRASH	CRASH	Arc / hop(x,y)
T45	Four equal squares forming one larger square	17%	FAIL	Composite
T46	Three equilateral triangles	CRASH	CRASH	Triangle / hop(x,y)
T47	Three squares in a triangular pattern	0%	FAIL	Composite
T48	Four nested squares with increasing side lengths	58%	FAIL	Composite / nesting
T49	Square with four quarter-circles at each corner	28%	FAIL	Arc + composite
T50	Square with a circle at each of its four corners	45%	FAIL	Arc + composite
T51	Hexagon with alternating large and small circles	61%	FAIL	Arc + polygon
T52	Octagon	36%	FAIL	Regular polygon
T53	Square, side length 100	30%	FAIL	Size-specific
T54	Six regular hexagons arranged in a ring	67%	PASS	Structured repeat
T55	Five concentric regular pentagons	64%	FAIL	Composite / nesting
T56	Three regular hexagons in a triangular pattern	33%	FAIL	Composite / polygon
T57	Large triangle, smaller triangle, large triangle in a row	30%	FAIL	Composite / triangle
T58	Two concentric circles	64%	FAIL	Arc / concentric
T59	Large square containing four smaller squares inside	55%	FAIL	Composite / nesting
T60	Four triangles of different sizes in a row	33%	FAIL	Composite / triangle
T61	Semicircle with horizontal diameter line	CRASH	CRASH	Arc / hop(x,y)
T62	Two circles intersecting, shared lens region	CRASH	CRASH	Arc / hop(x,y)
T63	Hexagon with semicircle on each of its six sides	54%	FAIL	Arc + polygon
T64	Square subdivided into four equilateral triangles	63%	FAIL	Composite / triangle

T65	Three semicircles in a triangular pattern	23%	FAIL	Arc / semicircle
T66	Large circle with four smaller circles inside	43%	FAIL	Arc / multi-circle
T67	Four semicircles arranged in a square pattern	14%	FAIL	Arc / semicircle
T68	Square with four corner semicircles (variant A)	41%	FAIL	Arc + composite
T69	Square with four corner semicircles (variant B)	24%	FAIL	Arc + composite
T70	Grid of nine equilateral triangles (3x3)	41%	FAIL	Composite / triangle
T71	Semicircle, radius 100	4%	FAIL	Arc / semicircle
T72	Square, side length 200 (variant A)	37%	FAIL	Size-specific
T73	Large equilateral triangle with smaller inside	CRASH	CRASH	Triangle / sqrt import
T74	Square, side length 200 (variant B)	80%	PASS	Size-specific
T75	Square with inner square rotated 45 degrees	36%	FAIL	Composite / orientation

All 75 tasks were evaluated. No tasks are missing. CRASH = Scala compilation failure (no NSS computed): T32, T44, T46, T61, T62 used hop(x,y); T73 used sqrt(3) without import. FAIL T26 scored exactly 65% — at the threshold but counted as FAIL. Size-specific tasks T72/T74 share the same query but different GT code due to archive indexing inconsistency. NSS threshold for PASS: 65%. Scores rounded to nearest integer percent.*